

Yuchen Wang

✉ yuchenwang0303@gmail.com 📍 Mountain View, CA 🔗 yuchenwang3.github.io 🐙 yuchenwang3 🌐 yuchen3

Education

University of Illinois Urbana-Champaign , Computer Science	Champaign, IL, USA 2025 – 2027
Peking University , Intelligent Science and Technology (Artificial Intelligence) • Zhi Class (Elite Program), Top 10%	Beijing, China 2021 – 2025

Experience

Alibaba U.S. , Research Intern — Agentic LLM Post-Training	Sunnyvale, CA, USA May 2026 – Dec 2026 8 months
- Contributed to Occamy-1.0 with Alibaba's Accio Team, focusing on execution-grounded data and post-training infrastructure. The model builds on Qwen3.6-35B-A3B with Marathon (SFT + HDPO) and Sprint (SFT) experts, model soup, and post-merge SAO. - Built a verifier-gated, multi-harness data and training pipeline with immutable provenance, token-exact proxy/TiTO replay, state reconstruction, context-segment lineage, and loss masks. Under the matched Claw-Eval T/C protocol, the model uses 19.5% fewer tokens per trajectory and 46.4% less trace wall time than the starting checkpoint; timeout rate falls from 9.88% to 2.18%. - The latest technical report records 82.2 Claw-Eval average (71.4 Pass³), 49.16 WildClaw-Bench, and 27.6% AutomationBench strict pass rate.	
Freedo Technology , Research Intern — 3D Vision and Generative AI	Beijing, China May 2025 – Aug 2025 4 months
Built an end-to-end depth-conditioned ControlNet pipeline for 3D reconstruction from noisy point clouds, improving geometric fidelity from 32.7% to 85.3%.	

Open Source Projects

Open-Source LLM Systems Engineering

Upstream reliability and performance work across NVIDIA NeMo, Megatron-LM, vLLM, vLLM/Vime, SGLang, verl, and ModelScope.

- Reliability and correctness: Muon and hybrid Mamba/MoE fixes, sequence packing, NCCL startup, pickle-safe cross-process HTTP failures, non-finite RL logprobs, GDN/Muon clipping and routing, skipped-tensor weight checks, and authoritative FSDP strategy.
- Performance and scale: vLLM #54699 halved TP2 peak allocation from 7.88 to 3.94 GiB and enabled real 120B load, generation, and 3/3 sleep-wake cycles on 8×B200; NeMo RL #3943 delivered 4.44–5.31× controlled Ray transfer speedup with 0.1–0.3 MB driver RSS deltas; Megatron-LM work fused GatedDeltaNet normalization and added Mamba recompute.

Projects

CineFlow — Dependency-Driven Parallel Video Generation

The first dependency-driven video-diffusion inference system, built around semantic DAGs, runtime trajectory fusion, online edge pruning, and critical-path scheduling without retraining.

- Analyzed 1K+ MovieGenBench prompts; more than 68% exhibit non-linear dependencies.
- Across Wan2.2-5B, CogVideoX-5B, and HunyuanVideo on 8×H100, achieved 1.7–5.5× end-to-end speedup, 1.30–2.02× lower P90 latency, and 5.4–17.3% higher VBench overall; pruning added nearly 7% throughput and improved visual coherence by 6.7–14.6%.

Dynamic Prefill Optimization via Adaptive Online Packing

AIMD control with live p95 TTFT feedback and greedy/DP prompt packing.

- Achieved up to 20% lower TTFT on production-style DynamoLLM traces.

FlashAttention-style CUDA Optimization for GPT-2

Tiled online softmax, memory-efficient attention, and kernel fusion.

- Reduced HBM traffic by roughly 10× and achieved up to 9% end-to-end speedup.

Reinforcement Learning for Legal Reasoning

Advised by Prof. Yansong Feng at Peking University; Zero-RL → distilled-CoT SFT → GRPO for structured legal multiple-choice reasoning.

- Reached 57.6% accuracy, outperforming SFT-only and RL-only baselines.

Skills

- Python, C++, CUDA, PyTorch, Ray, FSDP, 3D vision, point clouds, ControlNet, diffusion transformers, Megatron-LM, NeMo RL, vLLM, SGLang, distributed training, and profiling